

Part I — Leave-one-out stability of $\hat{\theta}(\mathcal{D})$

research/formal (working)

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Depends on `common_setup.tex`. Selected object = $\hat{\theta}$ from (2) only.

1 Stability notions (candidates)

Definition 1 (Exact LOO stability). $\hat{\theta}$ is exactly LOO-stable if for all i , $\hat{\theta}(\mathcal{D}) = \hat{\theta}(\mathcal{D}^{(i)})$ almost surely.

Definition 2 (Bounded LOO deviation). For a metric d on the hyperparameter space Θ ,

$$\Delta_{\text{LOO}}(\mathcal{D}) := \max_{1 \leq i \leq n} d(\hat{\theta}(\mathcal{D}), \hat{\theta}(\mathcal{D}^{(i)})).$$

$\hat{\theta}$ is (ε) -LOO stable on an event E if $\Delta_{\text{LOO}}(\mathcal{D}) \leq \varepsilon$ on E .

Definition 3 (Expected LOO stability). $\hat{\theta}$ is (ε) -stable in expectation if $\mathbb{E}[\Delta_{\text{LOO}}(\mathcal{D})] \leq \varepsilon$.

Definition 4 $((\eta, \tau, \nu)$ -stability of selection — Zrnic–Jordan Def. 2). A (possibly randomized) map $A : \mathcal{D} \mapsto \hat{\theta}$ is (η, τ, ν) -stable w.r.t. the data law if there exists an oracle A_0 (may depend on the law, not on the realized $\mathcal{D}$) such that with probability at least $1 - \nu$ over $\mathcal{D}$, the conditional law of $A(\mathcal{D})$ is (η, τ) -indistinguishable from A_0 in max-divergence: $\mathbb{P}\{A(\mathcal{D}) \in O \mid \mathcal{D}\} \leq e^\eta \mathbb{P}\{A_0 \in O\} + \tau$ for all measurable O (see digest `digest_zrnic-jordan_stability.md`, PDF Defs. 1–2).

Remark 1 (LOO $\neq$ Def. 4). Part I of the writeup asks leave-one-out movement of $\hat{\theta}$. Zrnic–Jordan stability is distributional indistinguishability of the selection law, typically enforced by randomizing selection. Both are tracked: LOO metrics above feed structural analysis of (2); Def. 4 feeds Part II validity certificates. Do not conflate them.

2 Assumptions on $\hat{f}, \hat{g}$ (W3)

Assumption 1 (Measurable dependence). $\hat{f}(\cdot; \theta)$ and $\hat{g}(\theta)$ are determined by $\mathcal{D}$ (filtration already in (2)).

Assumption 2 (Candidate regularity — OPEN). State per lemma, with warrant:

- Lipschitz / strong convexity of $\hat{f}$ in θ ;
- constraint qualification for $\hat{g}$;
- uniqueness of $\hat{\theta}(\mathcal{D})$.

No regularity imported from CREDO/CREME unless re-derived for (2).

3 Target lemmas (stubs)

Lemma 1 (LOO bound under strong convexity — STUB). *Hypothesis TBD after literature digest. If $\hat{f}$ is μ -strongly convex in θ and the active constraint set is stable under LOO, then $\Delta_{\text{LOO}}(\mathcal{D}) \leq \frac{2}{\mu} L_{\text{LOO}}$ for an explicit LOO perturbation size L_{LOO} of the objective.*

Lemma 2 (Randomized selection reduces LOO — STUB). *W5 hook. There exists a randomized map $\tilde{\theta}(\mathcal{D}; \xi)$ such that $\mathbb{E}_{\xi}[\Delta_{\text{LOO}}(\tilde{\theta})] \leq \mathbb{E}[\Delta_{\text{LOO}}(\hat{\theta})]$ (or a high-probability analogue), without modifying $\mathcal{C}(\cdot; \mathcal{D}, \theta)$'s construction for fixed θ .*

4 Randomized versions of (2) for goal (iii)

Definition 5 (Noisy-objective selection). *Given noise ξ , set*

$$\tilde{\theta}(\mathcal{D}; \xi) \in \arg \min_{\theta} \{ \hat{f}(y; \theta) + \langle \xi, \theta \rangle : \hat{g}(\theta) \leq 0 \}.$$

Definition 6 (Soft near-minimizer selection). *For budget $\varepsilon > 0$, draw $\tilde{\theta}$ from the set $\{ \theta : \hat{f}(y; \theta) \leq \hat{f}(y; \hat{\theta}) + \varepsilon, \hat{g}(\theta) \leq 0 \}$ (or an exponential tilt on suboptimality). Symbols from CREDO ε -argmin are quarantined — this is a selection thickener only.*

Lemma 3 (Structured vs isotropic — STUB). *If ξ is supported on the sensitivity subspace of the KKT system of (2) (active constraint normals / reduced Hessian), then for the same stability budget (η, τ, ν) the selection utility loss is weakly smaller than for isotropic $\xi \sim N(0, \sigma^2 I)$ under standard local quadratic approximations (pattern: `patterns_inverse_opt_structured_noise.md`).*

Problem 1 (Part I deliverable). *Prove usable forms of Lemmas 1–2, instantiate Defs. 5–6 with explicit (η, τ, ν) rates (Zrnic–Jordan design theorems as template), and feed into Part II. **Rates delivered:** see standalone `part_i_randomized_design.tex` (RNM / linear-Laplace $b = 2\varepsilon/\eta$; soft-argmin $\beta = \eta/(2\varepsilon)$; FW composition). Remaining: concrete $\varepsilon(n), \nu(n)$ for a chosen $(\hat{f}, \hat{g})$.*